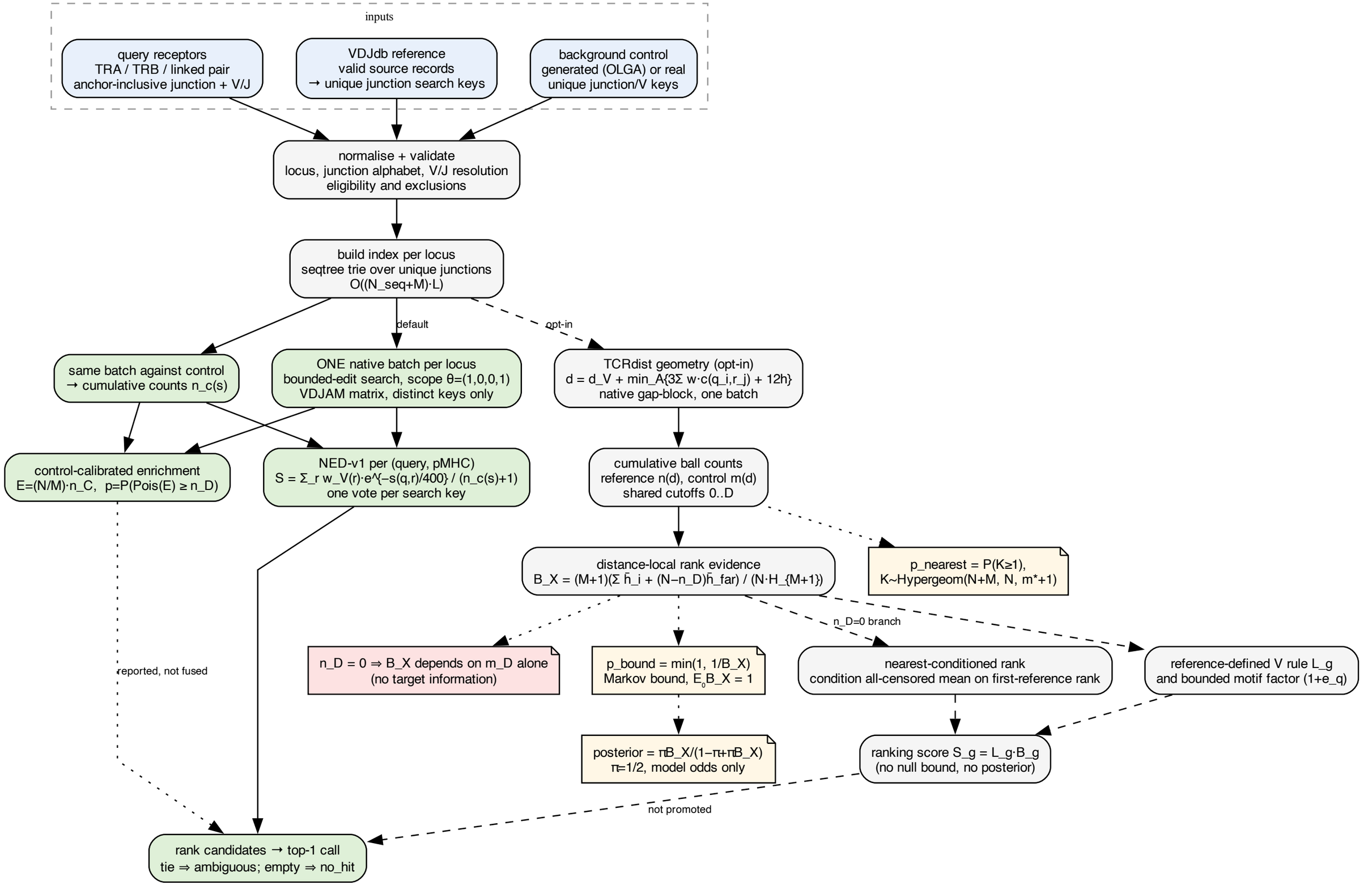
inputs
query receptors
TRA / TRB / linked pair
anchor-inclusive junction + V/J
VDJdb reference
valid source records
→ unique junction search keys
background control
generated (OLGA) or real
unique junction/V keys
normalise + validate
locus, junction alphabet, V/J resolution
eligibility and exclusions
build index per locus
seqtree trie over unique junctions
O((N_seq+M)·L)
default
opt-in
same batch against control
→ cumulative counts n_c(s)
ONE native batch per locus
bounded-edit search, scope θ=(1,0,0,1)
VDJAM matrix, distinct keys only
TCRdist geometry (opt-in)
d = d_V + min_A{3Σ w·c(q_i,r_j) + 12h}
native gap-block, one batch
control-calibrated enrichment
E=(N/M)·n_C, p=P(Pois(E) ≥ n_D)
NED-v1 per (query, pMHC)
S = Σ_r w_V(r)·e^{−s(q,r)/400} / (n_c(s)+1)
one vote per search key
cumulative ball counts
reference n(d), control m(d)
shared cutoffs 0..D
distance-local rank evidence
B_X = (M+1)(Σ h̄_i + (N−n_D)h̄_far) / (N·H_{M+1})
p_nearest = P(K≥1),
K~Hypergeom(N+M, N, m*+1)
n_D=0 branch
n_D = 0 ⇒ B_X depends on m_D alone
(no target information)
p_bound = min(1, 1/B_X)
Markov bound, E0B_X = 1
nearest-conditioned rank
condition all-censored mean on first-reference rank
reference-defined V rule L_g
and bounded motif factor (1+e_q)
reported, not fused
posterior = πB_X/(1−π+πB_X)
π=1/2, model odds only
ranking score S_g = L_g·B_g
(no null bound, no posterior)
not promoted
rank candidates → top-1 call
tie ⇒ ambiguous; empty ⇒ no_hit